Year 13 Further Maths - Rank correlation

Similar questions - Spearman's coefficient and hypothesis tests

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Six runners A, B, C, D, E, F take part in two different races. Spearman's rank correlation coefficient between the orders of finishing is calculated and a test for positive agreement is carried out at the 5% significance level (critical value 0.8286 for $n = 6$).

(a) State suitable hypotheses. **[1]**

(b) Find the largest value of $\sum d^2$ for which the result of the test is to reject the null hypothesis. **[3]**

(c) In the first race the order of finishing was A, B, C, D, E, F. In the second race, three of the runners finished in the same positions as in the first race, and the result of the test is to reject the null hypothesis. Find a possible order for the second race. **[3]**

2 Two magazines give ratings to six cameras. A student wishes to test whether there is agreement between the opinions of the magazines.

Camera	A	B	C	D	E	F
Magazine I	62	71	78	90	85	68
Magazine II	30	45	52	60	66	38

(a) Give a reason why the student might choose Spearman's rank correlation coefficient rather than Pearson's. **[1]**

(b) Calculate Spearman's rank correlation coefficient. **[4]**

(c) Carry out a test at the 5% significance level (critical value 0.8286 for a 1-tail test with $n = 6$). **[4]**

(d) The Spearman coefficient between Magazine I and a third magazine has the same size as your answer to (b) but the opposite sign. Complete the ranking of the third magazine. **[1]**

3 The numbers of nine items sold in two supermarkets in a week are given below.

Item	1	2	3	4	5	6	7	8	9
Supermarket A	17	28	41	43	62	69	75	93	115
Supermarket B	37	7	18	12	47	29	58	42	40

A researcher wants to test for association between the numbers sold. It is known that the data for B may be wrong by as much as 2 items in either direction.

(a) Explain why Spearman's rank correlation coefficient might be preferred to Pearson's. **[2]**

(b) Carry out the test at the 5% significance level (2-tail critical value 0.6833 for $n = 9$). **[8]**

Question 1

(a) H_0: there is no agreement between the orders in the two races; H_1: there is positive agreement

(b) $r_s = 1 - \dfrac{6\sum d^2}{6 \times 35} \geqslant 0.8286 \Rightarrow \sum d^2 \leqslant 6.0$. Largest value $\sum d^2 = 6$

(c) Keeping A, B, C in place and moving the others cyclically, the order A, B, C, E, F, D has $d = 0, 0, 0, 2, 1, 1$ (for D, E, F in the first-race order), so $\sum d^2 = 6$: allowed, and exactly three runners (A, B, C) finish in the same positions. (Any other order with exactly three unchanged has $\sum d^2 \geqslant 6$.)

Question 2

(a) The ratings are not necessarily from a bivariate normal population, or only the order of preference matters

(b) Ranks I: $1, 3, 4, 6, 5, 2$; ranks II: $1, 3, 4, 5, 6, 2$. d: $0, 0, 0, 1, -1, 0$, $\sum d^2 = 2$. $r_s = 1 - \dfrac{6 \times 2}{6 \times 35} = 0.943$

(c) H_0: no association; H_1: positive association. $0.943 > 0.8286$, so reject H_0: there is evidence of agreement

(d) Reverse the ranks of II ($7 - $rank) so that r_s changes sign: ranks $6, 4, 3, 2, 1, 5$ for A to F

Question 3

(a) Spearman uses only ranks, so a modest error in the values of B is unlikely to change the ranks much; it also does not require bivariate normality

(b) H_0: no association; H_1: there is association. Ranks A: $1 \ldots 9$; ranks B: $5, 1, 3, 2, 8, 4, 9, 7, 6$. $\sum d^2 = 48$, $r_s = 1 - \dfrac{6(48)}{9 \times 80} = 0.600$. Critical value 0.6833; $0.600 < 0.6833$ so there is insufficient evidence to reject H_0: no evidence of association